

Advanced Algorithms: Homework 1 Solutions

Due on Jan. 24, 2024 at 11:59pm EST

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As stated in the syllabus, unauthorized use
of previous semester course materials is
strictly prohibited in this course.

Exercise 1

(KT Chapter 1, Exercise 1) True or false? (If true, give a brief explanation, if false, give a counterexample): In every instance of the Stable Matching Problem, there is a stable matching containing a pair (m, w) such that m is ranked first on the preference list of w and w is ranked first on the preference list of m .

Solution

This is false. Here is an example with 4 people: m, m' and w, w' . Say m 's list is $[w, w']$, m' 's list is $[w', w]$. However, w 's list is $[m', m]$, and w' 's list is $[m, m']$.

There are two possible pairings, and both are stable. Either both men get their first choice and both women get their second choice, or vice-versa. In neither pairing is there a marriage where both partners rank the other first on their preference lists.

Exercise 2

(KT Chapter 1, Exercise 2) True or false?

Consider an instance of the Stable Matching Problem in which there exists a man m and a woman w such that m is ranked first on the preference list of w and w is ranked first on the preference list of m . Then in every stable matching S for this instance, the pair (m, w) belongs to S .

Solution

This is true. Assume otherwise - say there is a stable matching M where w is not matched to m , and m is not matched to w . Since each is first on the other's list, and they are currently matched to someone who is not first on their list, (m, w) is a rogue pair. Thus M isn't stable - contradiction!

Exercise 3

(KT Chapter 1, Exercise 3) There are many other settings in which we can ask questions related to some type of "stability" principle. Here's one, involving competition between two enterprises.

Suppose we have two television networks whom we'll call A and B . There are n prime-time programming slots, and each network has n TV shows. Each network wants to devise a schedule - an assignment of each show to a distinct slot - so as to attract as much market share as possible.

Each show has a fixed rating, which is based on the number of people who watched it last year; we'll assume that no two shows have exactly the same rating. A network *wins* a given time slot if the show that it schedules for that time slot has a larger rating than the other network schedules for that time slot. The goal is to win as many time slots as possible.

We'll say a pair of schedules (S, T) is *stable* if neither network can unilaterally change its own schedule and win more time slots.

For every set of TV shows and ratings, is there always a stable pair of schedules? Resolve this question by doing one of two things:

- (a) give an algorithm that, for any TV shows and associated ratings, produces a pair of stable schedules; or
- (b) give an example of TV shows and ratings for which there is no stable pair of schedules.

Solution

Unlike stable matching, there isn't always a stable schedule assignment. Say one network A has shows with ratings 1, 3, and the other network B has shows with ratings 2, 4. If B had it's choice, it would put 2 at the same slot as 1, and 4 at the same slot as 3, so that it would win both time slots. But this isn't stable, as A would swap it's schedule so that 1 is against 4, and 3 is against 2, and each network would win one time slot. This isn't stable either, as with this starting configuration, B would swap it's shows to win both time slots.

Exercise 4

Give an input to the Stable Marriage Problem with n men and n women that has a unique stable pairing. Now give an input with n men and n women that has an exponential number (in n) of stable pairings.

Solution

- One situation was presented in class: every man and every woman have the same preference list. $[m_1, m_2, \dots]$ for each woman, and $[w_1, w_2, \dots]$ for each man. Then, the only stable matching is $(m_1, w_1), (m_2, w_2), \dots$. Proof: Assume there is another matching M . Consider the smallest i so that m_i isn't matched with w_i . Then each of these is matched with someone lower on their preference list, and thus (m_i, w_i) is a rogue pair - contradiction.

Another situation with a unique stable matching is the following: say for all i , each person m_i has w_i as their ideal mate, and each w_i has m_i as their ideal mate. Then the matching $(m_1, w_1), (m_2, w_2), \dots$ is stable, since every person is paired with their ideal mate. Considering problem 2, this is the only possible stable matching. Note that it doesn't matter what all the other preferences are.

- There are many possible ways to get an exponential number of stable matchings, but here is one.

Say n is even, divide all the men and all the women into groups of 2. Label the men $m_1, m'_1, m_2, m'_2, \dots, m_{n/2}, m'_{n/2}$, and do the same for the women.

For each pair, we use a setup like the written solution from problem 1: Let each m_i 's list be $[w_i, w'_i, \dots]$, m'_i 's list be $[w'_i, w_i, \dots]$, and let each w_i 's list be $[m'_i, m_i, \dots]$, and w'_i 's list be $[m_i, m'_i, \dots]$. All other preferences after these first two won't matter.

For each i , say that we either have pairing $[(m_i, w_i), (m'_i, w'_i)]$, or pairing $[(m_i, w'_i), (m'_i, w_i)]$. We can see that this is stable because

- Everyone is paired with either their first or second choice, so there are no rogue pairs between some m_i and some w_j , $j \neq i$.
- No one can move unilaterally within their pair. As in the solution to problem 1, this is stable.

We can select either pair for each i without affecting any of the others. Thus there are at least $2^{\frac{n}{2}}$ stable matchings. (Without much more effort, you can show that these are the only stable matchings - but this is unnecessary).

Exercise 5

Let $[m]$ denote the set $\{0, 1, \dots, m-1\}$. For each of the following families of hash functions, say whether it is 2-universal or not, and specify how many random bits are needed to sample a hash function from the family, and then justify your answer.

- $H_a = \{h(x_1, x_2) = a_1x_1 + a_2x_2 \pmod{m} \mid a_1, a_2 \in [m]\}$, where m is some fixed prime. (Note that for each $h \in H_a$ we have $h : [m]^2 \rightarrow [m]$, i.e., it maps a pair of integers in $[m]$ to a single integer in $[m]$.)

- (b) $H_b = \{h(x_1, x_2) = a_1x_1 + a_2x_2 \pmod{m} : a_1, a_2 \in [m]\}$, where $m = 2^k$ is some fixed power of two.
- (c) The set of all functions $f : [m] \rightarrow [m-1]$.

Solution

- (a) This is a 2-universal family. We use two important facts from number theory that we also used in class.

- Fact 1: Let a be uniformly chosen from $[m]$. Then for ANY fixed x , $a + x \pmod{m}$ is uniformly distributed in $[m]$.
- Fact 2: Let a be uniformly chosen from $[m]$, where m is prime. Then for any fixed $x \neq 0$, $a \cdot x \pmod{m}$ is uniformly distributed in $[m]$.

This hash family is 2-universal if

$$\forall (x_1, x_2) \neq (y_1, y_2), [\Pr[f(x_1, x_2) = f(y_1, y_2)] \leq 1/m].$$

Notice that the $\forall$ is outside of the $\Pr$, so we *first* fix $(x_1, x_2) \neq (y_1, y_2)$, *then* we consider the Probability over all a_1, a_2 .

The event $f(x_1, x_2) = f(y_1, y_2)$ means that $a_1x_1 + a_2x_2 = a_1y_1 + a_2y_2 \pmod{m}$. With some algebra, it means $a_1(x_1 - y_1) + a_2(x_2 - y_2) = 0 \pmod{m}$. Since $(x_1, x_2) \neq (y_1, y_2)$, we know at least one of $x_1 \neq y_1$ or $x_2 \neq y_2$. Without loss of generality, assume that $x_1 \neq y_1$. Then, $(x_1 - y_1)$ is non-zero. Thus, $a_1(x_1 - y_1) \pmod{m}$ is distributed uniformly in $[m]$ by Fact (2). The other term $a_2(x_2 - y_2) \pmod{m}$ is something independent of a_1 . No matter what it $a_2(x_2 - y_2)$ happens to be, $a_1(x_1 - y_1) + a_2(x_2 - y_2) \pmod{m}$ is uniformly distributed in $[m]$ by Fact (1). Thus the the Probability that this is 0 is $1/m$.

A hash function in this family is determined by a_1, a_2 , each of which are $\in [p]$. Thus we need $2\lceil \log(p) \rceil$ bits.

- (b) This is NOT a 2-universal family. Now that $m = 2^k$ is not Prime, we can no longer use Fact (2) in our Proof above. Consider $(x_1, x_2) = (0, 0)$ and $(y_1, y_2) = (2^{k-1}, 0)$. Now, the event $f(x_1, x_2) = f(y_1, y_2)$ can be boiled down with some algebra to $a_1 2^{k-1} = 0 \pmod{2^k}$. If a_1 is a multiple of 2, then $a_1 2^{k-1}$ is a multiple of 2^k . Exactly half of the elements of $[2^k]$ are multiples of 2. Thus for any $k > 1$, $\Pr[f(0, 0) = f(2^{k-1}, 0)] = \frac{1}{2} > \frac{1}{2^k}$.
- (c) This is a 2-universal family. In this family, each input $i \in [m]$ has $m-1$ possible choices to map to. We can sample from this set in the following natural way - for each element $i \in [m]$, pick an element in $[m-1]$ uniformly at random, independently of our choices for all other $j \in [m]$. Therefore, for any fixed $x, y \in [m]$, $f(x)$ and $f(y)$ are independent, and $\Pr[f(x) = f(y)] = \frac{1}{m-1}$. Since our target set has size $m-1$, this is a 2-universal family.

We need to choose one element in $[m-1]$ for each element in $[m]$. Thus we need $m\lceil \log(m-1) \rceil$ bits.